

Tilak Bhusal

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Summary

Energy Engineer with a strong foundation in Mechanical and Industrial Engineering, specializing in energy efficiency, decarbonization, and sustainability. Experienced in conducting industrial and commercial energy audits, identifying cost-effective efficiency opportunities, and supporting data-driven decarbonization strategies. Background includes advanced analytics, machine learning, and advanced regression modeling using Python and R. Committed to driving impactful, equitable, and sustainable energy solutions across buildings and industry.

Education

West Virginia University, Morgantown, USA

2024 – 2025

M.S., Industrial Engineering, **GPA:** 3.88/4.0

Tribhuvan University, Pulchowk, Nepal

2018 – 2023

B.S., Mechanical Engineering, **GPA:** 3.75/4.0

Research Interests

Energy Efficiency, Energy Policy, Energy Economics, Decarbonization, Sustainability, Climate Change

Peer-Reviewed Publications

1. **Bhusal, T.**, Timilasina, S., Nimbarte, A., & Choudhury, A. (2025). *A Study on the Effect of Socioeconomic and Demographic Factors on Energy Consumption in Residential, Commercial, and Industrial Settings: A Case Study of West Virginia*. Proceedings of the West Virginia Academy of Science, 97(1).
2. Timilsina, S. S., **Bhusal, T.**, Nimbarte, A., & Choudhury, A. (2025). *Analysis of Socio-Demographic, Pollution, And Hazard Risk Factors Affecting Life Expectancy In West Virginia: A Multilevel Regression Approach: Factors Affecting Life Expectancy In West Virginia*. Proceedings of the West Virginia Academy of Science, 97(1).
3. Subedi, N., Pandey, A., & **Bhusal, T.** (2022). Study of Motorcycle Rear Suspension Behavior with Length of Swing Arm and its Inclination Angle. *VETOMAC*.

Work in Progress

1. **Bhusal, T.**, & Choudhury, A. (2026). *Determinants of Energy Consumption in Residential, Commercial, and Industrial Sectors: A Systematic Literature Review*. [Submitted for Peer-Review]

Conference Presentation

1. WVU Pollution Prevention Conference (2025)– *Empowering Industrial Energy Savings: Methods, Tools, and Impact of Energy Audits*
2. All Voices as One (2024) – *Electricity and Natural Gas Consumption in West Virginia Across Residential, Commercial, and Industrial Sectors*.

Research Experience

Graduate Research Assistant

WVU Pollution Prevention Lab

Jan 2024 – Dec 2025

- Conducted over 10 comprehensive energy assessments across various industrial sectors in West Virginia as part of ASHRAE Level I, II, and III audits.
- Identified and quantified energy efficiency and pollution prevention opportunities in lighting, HVAC systems, motors, compressed air, steam, and process heating systems.
- Utilized tools like MEASUR, eQUEST, and Excel-based calculators for energy modeling, equipment analysis, and ROI estimation.

Thesis:

- A multilevel regression study on the impact of socioeconomic and demographic factors on energy consumption patterns across residential, commercial, and industrial sectors in U.S. counties.
- Analyzed large datasets using R and Python to uncover systemic disparities in energy access and usage, aiming to inform equitable energy policy.
- Authored two peer-reviewed research articles and one review paper.

Undergraduate Researcher

Center for Pollution Studies

2022 – 2023

- Led the team for a government-funded research on three key areas: study of gaseous emissions, sustainability, and environmental benefits of energy transition, and techno-economic analysis.
- Completed the project on the economic viability of fuel transition from gasoline to electricity in the context of Nepal

- Student Researcher** Incubation, Innovation and Entrepreneurship Center 2020 – 2021
- Conducted research and developed a bike ambulance trailer funded by the Global Challenge Research Fund to support the livelihoods of the remote area in Nepal.
 - Design and fabrication of a disinfectant robot funded by the Nepal Academy of Science and Technology (NAST) during the COVID-19 pandemic.

Professional Experience

- Engineer** Energy Sciences Jan 2026 – Present
- Conduct onsite commercial and industrial facility assessments to identify and quantify energy efficiency opportunities across HVAC, motors, compressed air, process heating, and lighting systems, supporting utility energy efficiency programs.
 - Perform detailed engineering analyses, including regression modeling, bin-weather calculations, and trending-based evaluations, to calculate and verify energy and cost savings for recommended energy efficiency measures.
 - Develop high-quality technical reports and client-ready presentations summarizing findings, energy savings, and decarbonization strategies; collaborate directly with facility managers and stakeholders to advance energy projects.
 - Work cross-functionally with technical teams to evaluate project feasibility, support program implementation, and train stakeholders on energy engineering fundamentals and best practices in decarbonization and sustainability.

- Mechanical Design Engineer (Freelancer)** Fiverr 2021 – 2023
- Worked as a mechanical design and simulation engineer and delivered work related to CFD, 2-Way FSI, and Structural Analysis using ANSYS.
 - Completed various CAD projects, created domains for simulation projects, and designed machine components using SolidWorks.

- Design Lead** Robotics Club, Pulchowk Campus 2019 – 2021
- Design and Fabrication of robot sub-assembly and mechanisms for ABU Robocon 2019 and 2020.
 - Design and fabrication of Mushroom Incubator which helps in growing mushrooms in limited space.
 - Hands-on experience with CAD, CFD, and sensors and actuators.

- Intern** SunFarmer Nepal Summer 2022
- Designed solar arrays, pumps, and motors, and created CAD drawings for solar-powered projects.
 - Assisted in system optimization, project planning, and implementing renewable energy solutions for rural and underserved communities.
 - Conducted site monitoring and inspection, assisting in business development and procurement.

Skills

Energy Systems: Energy System Analysis, Utility Bill Analysis, Energy Audits, Auditing Equipment, Sustainability, Decarbonization

Software & Tools: Python, R, MEASUR, EnergyPlus, MS Office Suite, ANSYS, SolidWorks, Jasp, Jamovi, Matlab, Additive Manufacturing, FEA, CFD, CAD

Personal & Professional Skills: Critical Thinking, Teamwork, Brainstorming, Problem-Solving

Research & Data Analytics: Quantitative Analysis, Data Analysis, Documentation, Technical Writing, Research Methodology

Research and Technical Projects

1. Quality Inspection Using Machine Learning 2024
2. Energy Analytics in Action: Heating Tunnel Project 2024
3. Techno-Economic and Environmental Benefit Study of Gasoline Motorcycle to Electric 2023
 - Design of technical parameters (drive, battery pack, and CAD modeling).
 - Study the environmental benefits of conversion with an analysis of energy consumption, local emissions, and carbon savings.
 - Perform a Techno-Economic viability of the project in the context of Nepal.
4. Solar Mini Grid Feasibility Study 2023
 - Evaluated the technical and economic viability of a solar mini-grid system in a remote village.
 - Assessed potential for decentralized energy solutions and sustainability impacts.
5. Energising Development: Integrated Approach for Sustainable Electric Transportation 2021
 - Funded by Global Challenge Research Fund for co-designing sustainable electric transportation

solutions to support local livelihoods and inclusive economic growth in rural South.

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| 6. Disinfection Using Robotics in Hospital and Quarantine Settings | 2021 |
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Workshops and Trainings

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| 1. Harvard Aspire Leaders Fellow (Leadership and Professional Development) | 2023 |
| 2. Energy Audit Training for Young Engineers, Center for Energy Studies | 2023 |
| 3. Energy Efficiency in Electric Motors, Boilers and Water Pumping System | 2023 |
| 4. Energy Efficiency and Energy Savings Potentials for Industries – Virtual Practical Training | 2022 |
| 5. Research Methodology Training Program | 2022 |

Professional Participation

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| • AEE East Conference and Expo 2025 | 2025 |
| • National Renewable Energy and Policy Symposium, EnergizeNepal, Kathmandu University | 2023 |
| • National Mechanical Engineering Seminar, SOMAES | 2020 |
| • National Mechanical Engineering Seminar, SOMAES | 2019 |
| • National Young Scientists Conference, RECAST | 2019 |

Fellowship and Awards

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| • AEE Travel Support | 2025 |
| • Fully Funded Graduate Fellowship with Tuition Waiver, West Virginia University | 2024 |
| • Ministry of Physical Infrastructure and Transport, Nepal – Research Grant | 2022 |
| • Winner, Energy Hackathon
Awarded for providing insights and research on the techno-economic analysis and environmental benefits of fuel transition in the context of Nepal. | 2023 |
| • Winner, 3D Modeling Competition
Best design for disinfection robotics using a microbiological approach. | 2021 |
| • Tokyo Electron
International award for designing the best kicking mechanism in ABU Robocon 2020. | 2020 |
| • Rohm Award and Nagase Award
Awarded after defeating India and Thailand in the preliminary rounds of the international robotics competition. | 2019 |
| • Winner, Design Competition
National award for the best design in the open category, focusing on creating energy-efficient pumping systems for residential buildings. | 2019 |

Certifications

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| • LEED Green Associate (GBCI). |
| • Certified Energy Auditor (CES). |
| • Engineer-in-Training (NCEES). |
| • Professional Engineer, Nepal Engineering Council. |
| • Certified Solidworks Professional (CSWP) – Dassault Systèmes |

Professional Development and Volunteering

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| • Member, Alpha Pi Mu (<i>Industrial Engineering Honor Society</i>) | 2025 |
| • Association of Energy Engineers(AEE), Student Member | 2025 |
| • WVU Core Arboretum Cleanup | 2025 |
| • Campus Cleanup - WVU AEE | 2025 |
| • Member, AEE WVU Chapter | 2024 |

Teaching and Mentoring

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| • Judge, Undergraduate Research Symposium, WVU | Fall 2024 |
| • Judge, Undergraduate Research Symposium, WVU | Spring 2024 |
| • Instructor, SolidWorks Training Workshops | 2022 |
| • Co-coordinator, Design Challenge and Ideathon events, ZENIX | 2021 |

References

Avishek Choudhury, PhD
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IMSE, West Virginia University
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